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CITIZENSHIP	USA	
EMPLOYMENT	- University of Michigan (Ann Arbor, MI, USA) <i>Aug. 2021 – Present</i> Assistant Professor Computer Science & Engineering - Wesleyan University (Middeltown, CT, USA) <i>Dec. 2020 – Aug. 2021</i> Postdoctoral Researcher	
EDUCATION	Northeastern University, Boston, MA <i>2014 – 2020</i> PhD in Computer Science, <i>Dec, 2020</i> Thesis: <i>A Semantic Foundation for Sound Gradual Typing</i> Advisor: Amal Ahmed Committee: Matthias Felleisen, Ronald Garcia, Daniel R. Licata, Peter Thiemann, Mitchell Wand Northwestern University, Evanston, IL <i>2009 – 2014</i> MS in Computer Science, <i>June 2014</i> BA in Computer Science and Mathematics, <i>June 2013</i>	
RESEARCH INTERESTS	Programming language design, semantics and implementation; verified software; gradually typed programming languages; parsing; compiler intermediate languages; type theory; category theory	
FUNDING	NSF, CAREER: Foundations and Verification of Parsers and Type Checkers, 2540652, PI: Max S. New, \$680,000 May 2026-Apr 2031 AFOSR, Mechanized Denotational Semantics using Synthetic Category Theory, FA9550-23-1-0760, PI: Max S. New, \$711,841 Sep 2023-Sep 2028	
PHD ADVISEES	Eric Giovannini <i>Fall 2021-Present</i> , PhD Candidate Steven Schaefer <i>Summer 2023-Present</i> , PhD Candidate Yuchen Jiang <i>Fall 2023-Present</i> , PhD Candidate Eric Bond <i>Fall 2023-Present</i> , PhD Candidate Jesse Slater <i>Fall 2024-Present</i> , PhD Candidate, Co-advised with Xinyu Wang	
UNIVERSITY SERVICE	University of Michigan Admissions Committee <i>Fall 2025-Present</i> Hosting Committee <i>Fall 2022-Winter 2025</i> Graduate Committee <i>Fall 2021-Winter 2022</i>	

PROFESSIONAL ACTIVITIES AND SERVICE	Program Co-chair with Guilhem Jabert Twelfth Workshop on Higher Order Programming with Effects (HOPE 2024)	<i>Fall 2024</i>
	Co-organizer with Jean-Baptiste Jeannin, Cyrus Omar, Xinyu Wang Midwest Programming Languages Symposium 2023	<i>Fall 2023</i>
	Program Co-chair with Daniel Hillerström Eleventh Workshop on Higher Order Programming with Effects (HOPE 2023)	<i>Fall 2023</i>
	Program Co-chair with Jeremy Gibbons Ninth Workshop on Mathematically Structured Functional Programming (MSFP 2022)	<i>April 2022</i>
	Program Co-chair with Sam Lindley Eighth Workshop on Mathematically Structured Functional Programming (MSFP 2020)	<i>April 2020</i>
	Invited Participant Shonan Meeting No. 146: Programming and Reasoning with Algebraic Effects and Effect Handlers	<i>March 2019</i>
	Dagstuhl Seminar 18201: Secure Compilation	<i>May 2018</i>
	Panelist NSF Proposal Reviewer, 2022, 2025	
	Panelist Programming Languages Mentoring Workshop at POPL 2019 Panel: Grad School and Beyond	<i>January 2019</i>
	Co-chair with Gabriel Scherer New England Programming Languages and Systems Symposium (Selection Committee May 2016, June 2017, August 2018)	<i>October 2016</i>
	Program Committee Member (Conference) PLDI 2026, POPL 2024, OOPSLA 2023, MFPS 2022, ICFP 2019	
	Program Committee Member (Workshop) TYPES 2025, HATRA 2021, HATRA 2020	
	Journal Reviewing for: ACM Transactions on Programming Languages and Systems (TOPLAS), Journal of Functional Programming (JFP), Logical Methods in Computer Science (LMCS)	
	External Conference Reviewer POPL, ICFP, LICS, FoSSaCs, LNCS, TOPLAS, OOPSLA	

PUBLICATIONS	Intrinsic Verification of Parsers and Formal Grammar Theory in Dependent Lambek Calculus Steven Schaefer, Nathan Varner, Pedro H. Azevedo de Amorim, Max S. New <i>Proceedings of the ACM on Programming Languages</i> Notions of Stack-Manipulating Computation as Relative Monads Yuchen Jiang, Runze Xue, Max S. New <i>Proceedings of the ACM on Programming Languages</i> Denotational Semantics of Gradual Typing using Synthetic Guarded Domain Theory Eric Giovannini, Tingting Ding, Max S. New <i>Proceedings of the ACM on Programming Languages</i> Gradual Typing for Effect Handlers Max S. New, Eric Giovannini, Daniel R. Licata <i>Proceedings of the ACM on Programming Languages</i> A Formal Logic for Formal Category Theory Max S. New, Daniel R. Licata <i>International Conference on Foundations of Software Science and Computation Structures</i> Gradual Type Theory Max S. New, Daniel R. Licata <i>Journal of Functional Programming</i> Call-by-name Gradual Type Theory Max S. New, Daniel R. Licata <i>Logical Methods in Computer Science</i> Graduality and Parametricity: Together Again for the First Time Max S. New, Dustin Jamner, Amal Ahmed <i>Proceedings of the ACM on Programming Languages</i> How to evaluate the performance of gradual type systems Ben Greenman, Asumu Takikawa, Max S. New, Daniel Feltey, Robert Bruce Findler, Jan Vitek, Matthias Felleisen <i>Journal of Functional Programming</i> Gradual Type Theory Max S. New, Daniel R. Licata, Amal Ahmed <i>Proceedings of the ACM on Programming Languages</i> Graduality from Embedding-projection Pairs Max S. New, Amal Ahmed <i>Proceedings of the ACM on Programming Languages</i> Call-by-name Gradual Type Theory Max S. New, Daniel R. Licata <i>International Conference on Formal Structures for Computation and Deduction</i> FabULous Interoperability for ML and a Linear Language Gabriel Scherer, Max S. New, Nick Rioux and Amal Ahmed <i>International Conference on Foundations of Software Science and Computation Structures</i> Fair Enumeration Combinators Max S. New, Burke Fetscher, Robert Bruce Findler, Jay McCarthy <i>Journal of Functional Programming</i>	PLDI 2025 OOPSLA 2025 POPL 2025 OOPSLA 2023 FoSSaCs 2023 JFP Vol 31, 2021 LMCS Vol 16, Issue 1, 2020 POPL 2020 JFP Vol 29, 2019 POPL 2019 ICFP 2018 FSCD 2018 FoSSaCS 2018 JFP Vol 27, 2017
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	Fully Abstract Compilation via Universal Embedding Max S. New, William J. Bowman, and Amal Ahmed <i>Proceedings of the ACM on Programming Languages</i>	ICFP 2017
	Oh Lord, Please Don't Let Contracts be Misunderstood (Functional Pearl) Christos Dimoulas, Max S. New, Robert Bruce Findler, Matthias Felleisen <i>ACM SIGPLAN Conference on Object-oriented Programming, Systems, Languages, and Applications</i>	ICFP 2016
	A Coq Library For Internal Verification of Running-Times Jay McCarthy, Burke Fetscher, Max New, Daniel Feltey, Robert Bruce Findler <i>International Symposium on Functional and Logic Programming</i>	FLOPS 2016
	Is Sound Gradual Typing Dead? Asumu Takikawa, Daniel Feltey, Ben Greenman, Max S. New, Jan Vitek, Matthias Felleisen <i>ACM SIGPLAN Symposium on Principles of Programming Languages</i>	POPL 2016
WORKSHOP TALKS	A Call-by-push-value Scheme Max S. New <i>Scheme and Functional Programming Workshop</i>	Scheme Workshop 2026
	Relative Monads in Call-by-push-value as an Abstraction of Stack-Based Effects Max S. New <i>Higher-order Programming with Effects</i>	HOPE 2022
	From Call-by-push-value to Stack-based TAL? Max S. New <i>Syntax and Semantics of Low-Level Languages</i>	LOLA 2019
	Every Program in Your Redex Model, in Order RacketCon 2013	September 2013
TEACHING	University of Michigan	
	◇ EECS 483, <i>Compiler Construction</i> Upper-level undergraduate compilers course	Fall 2021, Fall 2022, Fall 2023, Winter 2024, Winter 2025, Winter 2026
	◇ EECS 598, <i>Category Theory for Computer Scientists</i> Graduate-level course on category theory and programming language semantics	Winter 2022, Winter 2023, Fall 2025

INVITED TALKS	Notions of Stack-Manipulating Computation as Relative Monads IFIP Working Group 2.8 - Functional Programming	<i>February 2026</i>
	Intrinsic Verification of Parsers in Dependent Lambek Calculus IFIP Working Group 2.8 - Functional Programming	<i>May 2025</i>
	Compiling with Call-by-push-value Mathematical Foundations of Program Semantics 2023	<i>June 2023</i>
	Gradual Typing for Effect Handlers POPV Seminar, Boston University	<i>May 2023</i>
	A Type Theory for Formal Category Theory Tallinn Institute of Technology	<i>March 2023</i>
	A Type theory for Formal Category Theory LIX Proofs and Algorithms Seminar, École polytechnique	<i>October 2022</i>
	Type Theoretic Gradual Typing PL Club, University of Pennsylvania	<i>June 2019</i>
	A Type Theoretic Approach to Gradual Typing Principles of Programming Seminar, Carnegie Mellon University	<i>October 2018</i>
	Semantic Foundations for Gradual Typing Mathematical Foundations of Program Semantics 2018	<i>June 2018</i>
	Call-by-name Gradual Type Theory Northeastern PL Seminar	<i>April 2018</i>
	Retractions and Blame Northeastern PL Seminar	<i>December 2016</i>